SiMPL Electronic Batch Record — PI Sheet | **NEW (XStep-based EBR)**

AZD0543 — Virus Inactivation + Concentration & Diafiltration (2000 L, Process 1)

Material / Product	AZD0543 — Antibody-Drug Conjugate (DSP, Process 1, GPF, 2000 L)
Records covered	PN 8012441 Virus Inactivation (Low pH) + PN 8012475 Concentration & Diafiltration (Large Skid UF/DF)
MABR / MPR	MABR-0027570 (VI) | PN 8012475 (C&D)
Process Order No.	____________
Batch #	____________
Plant / Facility	GPF
Effective Date	____________

How to read this: the digital EBR is assembled from XStep building blocks. Each card shows the original MPR section it replaces ("Original BR: §___") and is tagged NEW (built new), VARIANT (a new variant of an existing SMPL XStep), or REUSE (reuses an existing DE1 100 SMPL XStep as-is — shown as a summary, no new mock-up). For the full line-by-line comparison (verbatim old text beneath each card) see the companion **"New vs Old"** PDF; traceability is in the RTM.

15 new + **8** variant + **14** reuse = **37** XSteps for both records.

XStep ↔ Original MPR Crosswalk

EBR Phase	XStep	Type	Original MPR Section
Process Order Information & Front Matter	SMPL: Order Header Details	R	Header
	SMPL: Signature Table	R	Personnel ID
	SMPL: Display BOM	R	§4
	SMPL: Room & Equipment Assign	R	§7 (VI 7.3-7.4 / C&D 7)
	SMPL: Additional Manufacturing Supplies	V	§4
	SMPL: Additional Solution Batches (Solution Switch)	V	§5
	SMPL: Process Notes & Global Limits (Acknowledgement)	V	§6
Reusable Building Blocks (instantiated across BOTH records)	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	V	VI 7.5 · C&D 7.4-7.6
	SMPL: Product Vessel Weigh	R	VI §7/8/15 · C&D §9/11/16/17/18
	SMPL: Calc — Three Columns (Value1 op Value2 = Result)	R	VI 7.12 / 8.10 / 14.15 · C&D §7/16/17
	SMPL: Multi-Variable Calculation (3-4 operand)	V	C&D §12/16/17
	SMPL: Product Mixing	R	VI §15 · C&D §9/16/17
	SMPL: Product Sampling & DLIMS Submission	R	VI §15/Att1 · C&D §8/14/16/17/Att3
	SMPL: Product pH / Conductivity / Temperature	V	VI 7.1/15.5 · C&D 8.12/14.11/17
	SMPL: Hold Time Table	V	VI §19 (Att3) · C&D §9/§25 (Att5)
	SMPL: Product Recirculation Worksheet	N	VI Att6 · C&D Att7
	SMPL: Transfer Tubing / Hosing Information	R	VI Att5 · C&D Att1
	SMPL: Product Storage & Cross-Record	R	VI 15.8 · C&D §9/§18
	SMPL: Instruction & Sign-off	R	VI §8/16/20 · C&D §11/20
	SMPL: Yield Calculations	R	VI Att4 · C&D Att8
	SMPL: Comments / Deviations Section	R	VI Att7 · C&D Att9
	SMPL: Non-Routine Sampling Record	R	VI Att2 · C&D Att4
Virus Inactivation — PN 8012441 (Low pH, 2000 L)	SMPL: VI Treatment Vessel Setup (Comprehensive)	N	§8.4

	SMPL: Iterative pH Titration Table (Acid / Base)	N	§10 / 12 / 14
	SMPL: Incubation Timing & Incubated Sample	N	§11 / 13
	SMPL: Starting / Transfer Temperature Decision	N	§7.14 / 9
	SMPL: Store VI Product & Hold-Time Link	N	§15.8
Concentration & Diafiltration — PN 8012475 (Large Skid UF/DF, 2000 L)	SMPL: Record CIPDS / Skid / Cassette Information	N	§7.2 / §11
	SMPL: Run Skid Recipe (Batch ID, Start, Setpoints)	N	§8/10/13/15/19
	SMPL: Calculate Minimum Membrane Surface Area	V	§7.7
	SMPL: Process Input Calculations	N	§12
	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	N	Att2 (§13-15)
	SMPL: Conditional Continued Diafiltration	N	§14
	SMPL: Recovery Calculations & Execution	N	§16
	SMPL: Dilution Decision & Calculations	N	§17
	SMPL: PFI Storage Vessel, SHC Filter & Store	N	§18
	SMPL: Post-Processing: Cassette Re-use & Sanitization	N	§19 / §20

Process Order Information & Front Matter

SMPL: Order Header Details REUSE

Original BR: Header

Product / PN, process order numbers, recorded by — standard SAP order header.

Reuses: **DE1 100: Display Process Order Header Data (AZ Components) / SXS: SIMPL Order Header**

SMPL: Signature Table REUSE

Original BR: Personnel ID

Personnel identification: print name, signature, initials for everyone executing the record.

Reuses: **DE1 100: SMPL Header: Signature Table**

SMPL: Display BOM REUSE

Original BR: §4

Bill of Materials — equipment (VI Treatment Vessel / UF-DF skid, meters, scale, mixer, FIT) and components / filters / bags with SAP consumption. Reuses the standard BOM display XStep.

Reuses: **DE1 100: Display BOM Material Table (AZ Components) / SMPL Header: Review BOM Details**

SMPL: Room & Equipment Assign REUSE

Original BR: §7 (VI 7.3-7.4 / C&D 7)

Record the process room and assign, up front, every instrument / equipment used in the record — pH / conductivity meter, thermometer, floor scale / balance, mixer, stir plate, pump — with a calibration-interval check (writing an equipment ID attests visual inspection + calibration). Downstream steps then pull from this assignment via a Get [Type] button (Get Scale / Balance, Get pH / Cond Meter, Get Thermometer, Get Mixer, Get Stir Plate, Get Pump) that populates that type's ID / Description (+ Cal Due for calibrated instruments). Reuses SMPL: Room/Equipment Assign + SMPL: Record Text Value.

Reuses: **DE1 100: SMPL: Room/Equipment Assign + SMPL: Record Text Value**

SMPL: Additional Manufacturing Supplies VARIANT

Original BR: §4

Instructions

If additional materials (filter, bag, assembly, etc.) are used during processing, verify the material is acceptable per Section 4 and record its information; document the reason for the change in the Comments Section. Enter the Part No. — input validation displays the Material Description. Enter the Batch No. — the Exp. Date auto-populates. Serial No. / Autoclave ID / Exp. are recorded manually. **SAP consumption posts automatically via the Goods Issue process message (Z_PICONS, movement type 261).**

Reuses: **SMPL: Additional Assembly + SMPL: Material Consumption (Z_PICONS Goods Issue)**

Data Input

#	MPR Step No.	Part No.	Material Description	Batch No.	Exp. Date	Serial No.	Autoclave ID	Autoclave Exp.	Performed By *
1									

+ Add Row

Witnessed By *

SMPL: Additional Solution Batches (Solution Switch) VARIANT

Original BR: §5

Instructions

When a process solution must be replaced with a different batch during processing, record the switch: indicate the applicable step, confirm the new solution is the same Part No. as the one replaced, and record Part No. / Batch No. (and/or process order) / Exp. Date. Document the reason in the Comments Section. Enter the Batch No.; the Exp. Date auto-populates.

Reuses: **SMPL: Solution Summary - Data Recording**

Data Input

#	Applicable Step No.	Part Number	Batch Number	Exp. Date	Performed By *
1					

+ Add Row

Witnessed By *

SMPL: Process Notes & Global Limits (Acknowledgement) VARIANT

Original BR: §6

Instructions

Global process notes and limits that apply throughout the record, e.g.: all activities at 15-27 °C; assume 1 L = 1 kg, 1 mL = 1 g unless otherwise noted; verify all solutions / raw materials / standards are within expiration before use; writing an equipment ID attests visual inspection and calibration check; all parameters are Targets / Alert Limits unless otherwise noted (contact supervisor / initiate deviation when alert limits are exceeded); NaOH is corrosive — wear a face shield (C&D). The operator acknowledges these notes; each individual limit is enforced at the step where its parameter is captured.

Reuses: **DE1 100: SXS: Text Instructions with Sign-off (DE2 903: Long Text Instructions)**

Reviewed By *

Reusable Building Blocks (instantiated across BOTH records)

SMPL: Incoming Product Information (Weight / Concentration / DLIMS) VARIANT

Original BR: VI 7.5 · C&D 7.4-7.6

Instructions

Record the incoming / starting product characterization for each vessel or bag: **tare weight** and **net weight / volume** (kg = L), the **SoloVPE protein concentration** result (g/L, sample tested per SOP-0107091) with the DLIMS project / sample numbers it is reported under. Values carry from the upstream MPR where applicable. One row per vessel / bag. (Covers VI 7.5 Affinity Product Information and C&D 7.4 Record VF Product Information.)

Reuses: **SMPL: Solution Summary - Data Recording**

Data Input

Vessel / Bag #	Tare Weight (kg=L)	Net Weight / Volume (kg=L)	SoloVPE Concentration (g/L)	DLIMS Project No.	DLIMS Sample No.	Performed By *
1						

+ Add Row

Witnessed By *

SMPL: Product Vessel Weigh REUSE

Original BR: VI §7/8/15 · C&D §9/11/16/17/18

Weigh a product vessel: press Get Scale / Balance to retrieve the scale assigned in Room & Equipment Assign — Scale / Balance ID + Description + Calibration Due Date auto-populate — then record gross weight; net weight computed (Gross − Tare, 1 kg = 1 L). Reuses the standard weight-pattern XStep.

Reuses: **DE1 100: SMPL: Record Scale Weight (Tare Weight / Select Vessel Type & Tare Weight variants)**

SMPL: Calc — Three Columns (Value1 op Value2 = Result) REUSE

Original BR: VI 7.12 / 8.10 / 14.15 · C&D §7/16/17

Two-operand calculation archetype (Value 1 [× ÷ + −] Value 2 = Result). Reused wherever a single two-input calc is needed — e.g. VI 7.12 Net Weight of Affinity product = Gross − Tare, membrane / volume calcs. Reuses SMPL: Calc Three Columns.

Reuses: **DE1 100: SMPL: Calc Three Columns**

SMPL: Multi-Variable Calculation (3-4 operand) VARIANT

Original BR: C&D §12/16/17

Instructions

Multi-operand calculation archetype (mixed × ÷ + − operators). Enter the measured operands; carried-forward and constant values are read-only; the result is computed. Used for the process-input, recovery and dilution calc chains. Shown: a three-operand example.

Reuses: **SMPL: Three Variable Calc (extended to 4 variables)**

Data Input

Value 1		Value 2		Value 3		Result	Performed By *
	×		+		=		

Verified By *

SMPL: Product Mixing REUSE

Original BR: VI §15 · C&D §9/16/17

Mix product for the required duration (e.g. ≥ 15 min) per the applicable SOP (LevTech / MagMix / tank): press Get Mixer to retrieve the assigned mixer (Mixer ID + Description auto-populate), record agitation RPM, start/end time, computed duration. Reuses the weight+timer pattern XStep.

Reuses: **DE1 100: SMPL: Mixing Time (or SIMPL: Record Mixing and Agitation)**

SMPL: Product Sampling & DLIMS Submission REUSE

Original BR: VI §15/Att1 · C&D §8/14/16/17/Att3

Remove the sample per Attachment 1/3 and SOP-0107097; aliquot & label per SOP-0107056; submit to the DLIMS storage location per SOP-0080295; stamp sampling date/time; record DLIMS project / sample numbers. Reuses SMPL: Sampling Record + SMPL: Sample Submission Chart.

Reuses: **DE1 100: SMPL: Sampling Record + SMPL: Sample Submission Chart**

SMPL: Product pH / Conductivity / Temperature VARIANT

Original BR: VI 7.1/15.5 · C&D 8.12/14.11/17

Instructions

Measure pH, conductivity and temperature of the product / effluent using the assigned offline instruments (per SOP-0080297 / SOP-0107117). Press **Get pH / Cond Meter** and **Get Thermometer** to retrieve the instruments assigned in Room & Equipment Assign — each ID / Description / Calibration Due Date auto-populates. Each result is range-validated against the step limits (carried from the recipe); the UoM shows beside each value. **Contact the supervisor/designee if a result is out of specification.**

Reuses: **SMPL: Solution Summary - Data Recording (range-validated variant)**

▸ Get pH / Cond Meter

pH / Cond Meter ID

pH / Cond Meter Description

pH / Cond Meter Cal Due

▸ Get Thermometer

Thermometer ID

Thermometer Description

Thermometer Cal Due

Data Input

#	Offline pH	UoM	Conductivity	UoM	Temperature	UoM	Performed By *
1		pH		mS/cm		°C	

+ Add Row

Witnessed By *

SMPL: Hold Time Table VARIANT

Original BR: VI §19 (Att3) · C&D §9/§25 (Att5)

Instructions

Track product hold time across storage moves. For each hold, select the storage condition, stamp storage start and end; RT and 2-8 °C durations are computed separately. Totals are computed — alert if RT total exceeds 72 h or the combined total exceeds 168 h (contact supervisor).

Reuses: **SMPL: Solution Summary - Data Recording (hold-time variant)**

Data Input

Hold #	Location	Storage Temp		Storage Start		Storage End	RT Hold (h)	2-8°C Hold (h)	Performed By *
1		2-8 °C	▸ Record		▸ Record				

+ Add Row

Total RT Hold (h)

Total 2-8°C Hold (h)

Total Combined Hold (h)

Witnessed By *

SMPL: Product Recirculation Worksheet NEW

Original BR: VI Att6 · C&D Att7

Instructions

Record each product recirculation used to mix the product when a mixer is not used (referenced from the mixing step). Stamp the start and end times per recirculation; the duration is computed. **No existing DE1 100 XStep matched a search for “recirculation” — built new** (a start/end/duration timer worksheet; closest relative is the SMPL: Mixing Time timer pattern).

Data Input

#	Recirculation Step	Start	Start Time	End	End Time	Duration (min)	Performed By *
1		▸ Start		▸ End			

+ Add Row

Witnessed By *

SMPL: Transfer Tubing / Hosing Information REUSE

Original BR: VI Att5 · C&D Att1

Record each transfer tubing / flex hose used: Part No., Batch No. (Exp. auto), autoclave cycle / expiry, hose serial no., cleaning batch ID / expiry. Reuses SMPL: Material Consumption / SMPL: Additional Assembly.

Reuses: **DE1 100: SMPL: Material Consumption / SMPL: Additional Assembly**

SMPL: Product Storage & Cross-Record REUSE

Original BR: VI 15.8 · C&D §9/§18

Record the initial storage condition (RT / 2-8 °C) and storage start date/time, and map values into the Hold-Time attachment / cross-record the storage linkage to the next MPR. Reuses the storage-record pattern.

Reuses: **DE1 100: SMPL: Solution Final Storage (cross-record write-back authored in MBR)**

SMPL: Instruction & Sign-off REUSE

Original BR: VI §8/16/20 · C&D §11/20

Reusable instruction-only / set-up shell (line & valve connections, dip-tube confirmation, N/A gates, product transfer between vessels, BPR review). The instruction text is authored per use in IV_INSTR; the operator reads it and signs. No data captured. Reuses Long Text Instructions (DE2 903).

Reuses: **DE1 100: SXS: Text Instructions with Sign-off (DE2 903: Long Text Instructions)**

SMPL: Yield Calculations REUSE

Original BR: VI Att4 · C&D Att8

Compute batch yield: % Yield = Final Product Mass ÷ Load Mass × 100 (masses carry from the weigh / concentration steps). Reuses SMPL: Yield Calculations.

Reuses: **DE1 100: SMPL: Yield Calculations**

SMPL: Comments / Deviations Section REUSE

Original BR: VI Att7 · C&D Att9

Free-text log for comments and documented deviations (step no. + initials/date). Reuses SXS: Phase Comments.

Reuses: **DE1 100: SXS: Phase Comments (no clean 'SMPL: Comments' — closest confirmed component)**

SMPL: Non-Routine Sampling Record REUSE

Original BR: VI Att2 · C&D Att4

Ad-hoc samples requested by MEMO: process section, contents, container, quantity, designation, DLIMS. Reuses SMPL: Non-Routine Sampling Record.

Reuses: **DE1 100: SMPL: Non-Routine Sampling Record**

Virus Inactivation — PN 8012441 (Low pH, 2000 L)

SMPL: VI Treatment Vessel Setup (Comprehensive) NEW

Original BR: §8.4

Instructions

Set up the VI Treatment Vessel. The vessel is either a single-use **bag** or a **tank**, plus (optionally) a vent filter and a required set of product filters. Use each **Required?** dropdown to activate only the sections that apply — a section set to *No* is deactivated and its fields are N/A. Component consumption posts via the Goods Issue process message (Z_PICONS, movement type 261) on the marked tables (per line). **Label** the vessel per SOP-0107056 (“AZD0543”, “VI Treatment Vessel”, batch/part no., initials, date).

Bag Required? * Yes / No — *activates / deactivates the section below*

Bag Information (if applicable) SAP Goods Issue · Z_PICONS mvt 261

Bag #	Part No.	Batch No.	Exp. Date	Performed By *
1				

+ Add Row

Mixer Drive Attached? * Yes / No

Top Base Attached? * Yes / No

Magnetic Clamp Attached? * Yes / No

Tank Required? * Yes / No — *activates / deactivates the section below*

Tank ID *

CIP Batch ID *

▸ Record

Date *

Time *

Pressure Test ID *

Pressure Test Date *

SIP Batch ID *

Vent Filter Required? * Yes / No — *activates / deactivates the section below*

Vent Filter Information (if applicable) SAP Goods Issue · Z_PICONS mvt 261

Part No.	Batch No.	Exp. Date

Product Filter Information (required) SAP Goods Issue · Z_PICONS mvt 261

Filter #	Part No.	Batch No.	Exp. Date	Autoclave Cycle No.	Autoclave Exp.	Performed By *
1						

+ Add Row

If the collection vessel is a bag, the tare weight is obtained prior to attaching the product filter.

▸ Get Scale / Balance

Scale / Balance ID	
Scale / Balance Description	
Scale / Balance Cal Due	
Tare Weight of VI Treatment Vessel (kg) *	
Filters attached when tare obtained *	Product filter / Vent filter / No filters (bag)
Recorded By *	
Witnessed By *	

SMPL: Iterative pH Titration Table (Acid / Base) NEW

Original BR: §10 / 12 / 14

Instructions

Iterative pH adjustment by incremental buffer addition. Record initial pH / conductivity / temperature and start time. Per addition: gross weight before & after (net added and running total computed), stamp the pH-sample time and record adjusted pH. Target per step — acid pH 3.5 (NOR 3.4-3.6), neutralization pH 7.2-7.6. **Base variant:** re-standardize the pH meter (pH 4.0/10.0) once pH ≥ 4.0. Buffer consumption posts via Goods Issue (Z_PICONS, 261) on the total net weight added.

▸ Get Scale / Balance

Scale / Balance ID

Scale / Balance Description

Scale / Balance Cal Due

▸ Get pH / Cond Meter

pH / Cond Meter ID

pH / Cond Meter Description

pH / Cond Meter Cal Due

▸ Get Thermometer

Thermometer ID

Thermometer Description

Thermometer Cal Due

Initial pH *

Initial Conductivity *

Initial Temperature (°C) *

Data Input

Addition #	Gross Wt Before (kg)	Gross Wt After (kg)	Net Added (kg)	Total Net (kg)		pH Sample Time	Adjusted pH	Performed By *
1					▸ Record	⏱		

+ Add Row

▸ Record

Final pH Confirmed Time *

Final pH

Final Conductivity *

Final Temperature (°C) *

▸ Record

Incubation Start Time *

Witnessed By *

SMPL: Incubation Timing & Incubated Sample NEW

Original BR: §11 / 13

Instructions

Time the low-pH incubation. The start time carries from the acid titration. Take a mid-incubation temperature check (bag) and an incubated sample (pH / conductivity / temperature); stamp the end time. Incubation duration is computed — target 75 min (NOR 60-240).

Incubation Start Time (from titration)

▸ Get Thermometer

Thermometer ID

Thermometer Description

Thermometer Cal Due

Mid-Incubation Temperature (°C) *

▸ Get pH / Cond Meter

pH / Cond Meter ID

pH / Cond Meter Description

pH / Cond Meter Cal Due

Incubated pH *

Incubated Conductivity *

Incubated Temperature (°C) *

▸ Record

Incubation End Time *

Incubation Duration (min)

Performed By *

Witnessed By *

SMPL: Starting / Transfer Temperature Decision NEW

Original BR: §7.14 / 9

Instructions

Conditional temperature check. Measure the product temperature and select the outcome: 18-25 °C → proceed to acid addition; < 18 °C → start warming (mixing section); > 25 °C → contact the supervisor. The selected branch drives which downstream sections are active.

Measured Temperature (°C) *

Decision (18-25 proceed / <18 warm / >25 supervisor) * Select

Performed By *

Witnessed By *

SMPL: Store VI Product & Hold-Time Link NEW

Original BR: §15.8

Instructions

Store the neutralized VI product and document storage information. Select the storage condition, stamp the storage start date/time, and map the values into Attachment 3 (Hold Times via the reusable Hold Time Table).

Storage Condition (RT / 2-8 °C) *	2-8 °C
	▸ Record
Date *	
Time *	
Performed By *	
Witnessed By *	

Concentration & Diafiltration — PN 8012475 (Large Skid UF/DF, 2000 L)

SMPL: Record CIPDS / Skid / Cassette Information NEW

Original BR: §7.2 / §11

Instructions

Record the UF/DF skid and membrane identity: Skid ID (DF-1244), CIPDS batch, Pellicon cassette lot / part, support-plate ID. Confirm the skid is within its post-sanitization storage window (≤ 30 days) — if exceeded, re-sanitize before use. **Label** the Retentate Vessel ("Retentate Vessel") and its manual valves (A–I) per SOP-0107056. Cassette / CIPDS component consumption posts via Goods Issue (Z_PICONS, 261).

Skid ID *

CIPDS Batch *

Cassette Lot / Part No. *

Support Plate ID *

Skid Storage ≤ 30 days? * Yes / No

▸ Get Scale / Balance

Scale / Balance ID

Scale / Balance Description

Scale / Balance Cal Due

Retentate Vessel Tare — Before Attach (kg) *

Retentate Vessel Tare — After Attach (kg) *

Performed By *

Witnessed By *

SMPL: Run Skid Recipe (Batch ID, Start, Setpoints) NEW

Original BR: §8/10/13/15/19

Instructions

Download and run the named skid recipe per SOP-0107111 (Prep-WFI Flush, Equilibration, Concentration 1, Diafiltration, Concentration 2, Recovery, Clean-WFI Flush, Sanitization). Enter the batch Name ID — format MPR PartNumber_Phase_BatchNumber_Date — and stamp the start time. Verify flow paths / setpoints match the recipe; contact area management on any discrepancy. One row per recipe run. Buffer (PN 8012555 equilibration / DF buffer), WFI and NaOH consumed by the recipe phases post via Goods Issue (Z_PICONS, movement type 261).

Data Input

Run #	Recipe / Phase	Batch Name ID		Start Date	Start Time	Performed By *
1			▸ Record			

+ Add Row

Witnessed By *

SMPL: Calculate Minimum Membrane Surface Area VARIANT

Original BR: §7.7

Instructions

Calculate the minimum membrane surface area required for the load and verify the installed area meets it: Total Grams Product ÷ Max Loading (g/m²) = Minimum Membrane Area (m²). Confirm installed area ≥ minimum. (Section 7 also records VF product information and the CIPDS/skid material consumption, which posts via Goods Issue (Z_PICONS, 261).)

Reuses: **SMPL: Calc Three Columns / Three Variable Calc**

Data Input

Total Grams Product (g)		Max Loading (g/m²)		Min Membrane Area (m²)	Performed By *
	÷		=		

Installed Membrane Area (m²) *

Installed ≥ Minimum?

Verified By *

SMPL: Process Input Calculations NEW

Original BR: §12

Instructions

Compute the process setpoints from the load: system capacity at 80 % / 40 %, Concentration 1 and Concentration 2 target volumes & weights, and the diafiltration permeate volume range. Inputs carry from the product-info and membrane steps; densities are step-specific constants; results are read-only. (Individual calcs reuse the Multi-Variable Calculation archetype.)

Starting Product Volume / Grams (from Step 7)

Target Conc 1 (g/L)

Conc 1 Target Volume (L)

Conc 1 Target Weight (kg)

Diafiltration Volumes (DV) *

DF Permeate Volume Range (L)

Target Conc 2 (g/L)

Conc 2 Target Volume (L)

Conc 2 Target Weight (kg)

Performed By *

Verified By *

SMPL: UF/DF Operation Monitoring Worksheet (Att 2) NEW

Original BR: Att2 (§13-15)

Instructions

Log operating parameters during Concentration 1 / Diafiltration / Concentration 2 at intervals (approx. every 20 min) using the same conductivity meter as set-up. Record TMP, feed / retentate / permeate pressures, permeate flow, conductivity and vessel weight; the max-pressure alarm must not be exceeded. Press **Get Conductivity Meter** to retrieve the meter — **it must match the meter used at set-up (Step 8.2)**.

▸ Get Conductivity Meter

Conductivity Meter ID

Conductivity Meter Description

Conductivity Meter Cal Due

Data Input

#		Time	TMP (bar)	Feed Press. (bar)	Retentate Press. (bar)	Permeate Flow (LMH)	Conductivity (mS/cm)	Vessel Weight (kg)	Performed By *
1	▸ Record	⏱							

+ Add Row

Witnessed By *

SMPL: Conditional Continued Diafiltration NEW

Original BR: §14

Instructions

At the end of the planned diafiltration, sample the permeate (pH / conductivity). If in range, proceed; if not, run additional diavolumes and re-sample. Repeat until in specification. Additional diafiltration buffer (PN 8012555) consumed posts via Goods Issue (Z_PICONS, 261).

▸ Get pH / Cond Meter

pH / Cond Meter ID

pH / Cond Meter Description

pH / Cond Meter Cal Due

Permeate pH *

Permeate Conductivity (mS/cm) *

Continue Diafiltration? * Yes / No

Additional Diavolumes (DV) *

Performed By *

Witnessed By *

SMPL: Recovery Calculations & Execution NEW

Original BR: §16

Instructions

Compute the recovery target and recover product to the Dilution Vessel. Choose the recovery method (manual flush vs recipe). Buffer used in recovery posts via Goods Issue (Z_PICONS, 261). Obtain and **label** the PFI mixing bag per SOP-0107056. Record the recovered volume / weight. (Volume/weight calcs reuse the Multi-Variable Calculation archetype.)

Target Recovery Volume (L)	
Recovery Method (Manual / Recipe) *	Select
	▸ Get Mixer
Mixer ID	
Mixer Description	
	▸ Get Scale / Balance
Scale / Balance ID	
Scale / Balance Description	
Scale / Balance Cal Due	
Recovered Weight (kg) *	
Recovered Volume / Concentration	
Performed By *	
Witnessed By *	

SMPL: Dilution Decision & Calculations NEW

Original BR: §17

Instructions

Based on the recovered concentration vs the PFI target range, decide whether dilution is required (target ~180 g/L PFI). If required, compute the dilution buffer amount; add buffer (Goods Issue, Z_PICONS 261), mix and sample. Obtain and **label** the "Intermediate Vessel" per SOP-0107056.

Recovered Concentration (g/L)	
Dilution Required? (Yes / No / Out-of-Range) *	Select
Target PFI Concentration (g/L)	
Dilution Buffer Amount (kg)	
Addition Method (Pump / Gravity) *	Select
	▸ Get Pump
Pump ID	
Pump Description	
	▸ Get Stir Plate
Stir Plate ID	
Stir Plate Description	
	▸ Get Scale / Balance
Scale / Balance ID	
Scale / Balance Description	
Scale / Balance Cal Due	
Final Concentration (g/L) *	
Performed By *	
Verified By *	

SMPL: PFI Storage Vessel, SHC Filter & Store NEW

Original BR: §18

Instructions

Obtain and label the PFI storage vessel and the SHC filter (record IDs / batch / expiry; consumption via Goods Issue, Z_PICONS 261). Transfer the diluted product through the SHC filter into the storage vessel, weigh, label and store; document hold times (Attachment 5).

PFI Storage Vessel ID *

SHC Filter Part / Batch / Exp. *

▸ Get Scale / Balance

Scale / Balance ID

Scale / Balance Description

Scale / Balance Cal Due

Net Weight / Volume (kg = L)

Storage Condition * 2-8 °C

▸ Record

Date *

Time *

Performed By *

Witnessed By *

SMPL: Post-Processing: Cassette Re-use & Sanitization NEW

Original BR: §19 / §20

Instructions

After processing, decide whether the cassette is re-used or discarded, then run the post-use Clean-WFI flush and NaOH sanitization / storage recipe (NaOH consumption via Goods Issue, Z_PICONS 261). Record the storage batch and start time.

Cassette Disposition (Re-use / Discard) * Select

NaOH Sanitization Batch *

Storage Batch *

▸ Record

Sanitization / Storage Start Time *

Performed By *

Witnessed By *